

# AVA: An Audio-based Virtual Aiming System for Accessible VR Shooting Games

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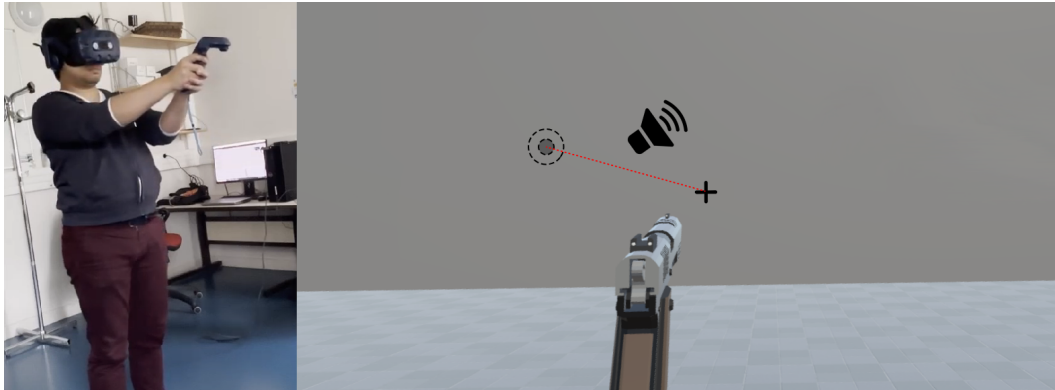


Figure 1: Left: Participant in a virtual reality experiment aiming at an invisible target using audio guidance from AVA. Right: VR scene with schematization of the target & central sound area (dotted circles), user's aim (cross), and sonified distance (red line).

## ABSTRACT

Video games provide a rich source of entertainment and social interaction, yet they remain largely inaccessible to blind and visually impaired individuals (BVI). AVA is an Audio-based Virtual Aiming system designed to guide BVIs in shooting video games using sonification methods in Virtual Reality (VR). Our current goal is to design sonification methods that simultaneously optimize the accuracy and speed of shooting tasks in VR games for BVIs. We are currently conducting two successive experiments comparing three sonification methods (Pitch, Tempo & Pitch and, Binary Pitch & Tempo), and two configurations: one exploring the impact of adding a sound at the center of targets, the other examining different sound spatialization approaches to facilitate target localization.

**Index Terms:** Sonification, Accessibility, Virtual Reality, Sport, Video Games, Shooting Task, Visual Impairment

## 1 INTRODUCTION

Virtual Reality (VR) accessibility is a major challenge in current research for the inclusion of people with disabilities. VR offers unique possibilities in terms of immersion, realistic simulation of complex environments and gestural interaction. Yet, despite these advances, blind and visually impaired people (BVI) remain largely excluded from VR experiences due to the lack of adapted accessibility solutions. Research has mainly focused on VR applications for BVIs in educational, assisted navigation, or training contexts in controlled and safe environments [3, 6].

In VR video games, user interaction relies heavily on vision and spatial perception, posing important barriers for BVIs. Im-

mersive mechanisms such as locating objects in space, aiming, or moving become inaccessible without effective sensory alternatives. Although mainstream video games are widely available, only a few are designed to accommodate vision impairments. Therefore, players must develop their own strategies to navigate inaccessible games, such as intentionally colliding with walls or using in-game objects as landmarks [4]. Research on video games for BVI players focuses on creating adapted frameworks and tools to enhance accessibility, modifying existing games, or designing audio games specifically tailored to their needs [5]. However, audio games mostly appeal to BVI players, which contrasts with the wishes expressed by the disabled gaming community to participate in the same activities as sighted gamers [1].

A promising way to overcome these barriers is the use of non-verbal sounds to complement and make visual or spatial information accessible to BVIs. Sonification [8] compensates for the absence of visual information and exploits users' sound localization abilities and auditory perception to create accessible interactions.

We created AVA, an Audio-based Virtual Aiming system designed to improve accessibility of shooting tasks in VR for BVIs with sounds. Using AVA, we explore sonification applied to First Person Shooter (FPS) VR games requiring high performance in terms of accuracy and speed. In a previous experiment [2], four sonification methods were evaluated to assist players with aiming tasks. The results indicated that while all methods achieved good accuracy, they lacked sufficient speed. We hypothesize that the presence of a contrasting sound at the center of the target might have encouraged participants to prioritize accuracy at the expense of speed, which is an essential criterion in FPS games. To test this hypothesis and enhance speed performance, we propose two new experiments. These experiments examine the influence of contrasting sounds at the target's center and assess how different spatialization techniques, which is a widely researched parameter in guidance systems for BVIs [7, 9], contribute to shooting performance.

## 2 AVA: AUDIO-BASED VIRTUAL AIMING

The AVA system was developed in Unity using SteamVR and PureData to assist BVI users in aiming at targets within a VR environ-

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ment. The system uses several angles to sonify different aspects of aiming, effectively guiding the user to point toward the target.

While the user aims with the controller, the accuracy is determined by calculating the angle between the gun-target vector and the gun's aiming vector. This angle is normalized and sent to a PureData module, a visual programming language for sound design and multimedia management in real time, via the LibPdIntegration plugin to generate audio feedback that represents the distance to the target's center. Additionally, the gun's aiming vector is projected onto both vertical and horizontal planes, enabling separate measurements of vertical and horizontal angles. These angles are used to sonify each axis independently, influencing distinct sound parameters to give users more information. Finally, sound is emitted from a sound source attached to the user. This mobile sound source enables the use of spatialized audio, allowing us to move the audio source according to the user's aim.

In a previous study [2], four sonification methods from the literature were evaluated with 24 sighted participants tasked with shooting at invisible targets using only sound feedback. To even some design elements across the sonification methods, a contrasting sound was added to all methods to indicate when the target's center was reached. Results showed comparable precision across methods, but the central sound may have biased results by encouraging participants to prioritize accuracy over speed, waiting for the cue before shooting. To address this, we designed a new experiment comparing sonification methods with and without central sounds (CS) to test two hypotheses: (1) removing CS will improve shooting speed but reduce accuracy, and (2) the absence of CS will benefit certain sonification methods more significantly than others.

Additionally, we developed an experiment investigating the effects of spatialization (*SPAT*). Previous research [7, 9] has shown that it is an effective guidance tool and an intuitive parameter to direct users toward a target due to its inherent spatial nature. Two spatialization approaches are tested: a binary method, indicating whether the target is left or right, and a continuous method, providing directional guidance from any angle. We hypothesize that (1) spatialization will enhance overall performances compared to non-spatialized methods, with (2) the continuous approach offering faster performance than the binary approach, while (3) the binary approach is predicted to be less mentally demanding for the user.

### 3 SONIFICATION METHODS & COMBINED PARAMETERS

Based on user feedback and performance outcomes, only three of the previously evaluated methods<sup>1</sup> in [2] were selected for further study to improve speed performances.

*Pitch Only* - This method is based on the pitch of the sound. The closer the aim is to the center of the target, the higher the pitch becomes.

*Tempo & Pitch* - This method is based on vertical pitch and horizontal tempo. On the vertical axis, the closer the aim is to the center, the higher the pitch. On the horizontal axis, the closer to the center, the faster the tempo.

*Tempo & Binary Pitch* - This method is based on vertical pitch and horizontal tempo. Above the target, there is a low note. Below the target, there is a high note. Similar to the previous method, on the horizontal axis, the closer the target is to the center, the faster the tempo becomes.

In addition to evaluating the sonification methods, we investigate two design choices to assess their impact on shooting speed for each method. The first experiment examines the inclusion of central sounds, while the second focuses on the incorporation of spatialization into these methods.

*Central Sound* - A few modifications have been added to the previous methods in the conditions with central sound. To contrast

with sounds around the target center, when the aiming reaches an angle  $\leq 1^\circ$  on the vertical axis, the note becomes medium. Similarly, on the horizontal axis, the sound becomes continuous. As a result, at the center of the target, a continuous medium note is heard.

*Spatialization* - Two spatialization methods are evaluated to assess their guidance efficiency in sonification methods. In the *binary approach*, the sound is heard opposite to the aim's position relative to the target. As a result, when the shooter aims to the right of the target center, they hear the sound at constant volume in his left ear, and vice versa. In the *3D continuous approach*, the sound is emitted by a sound source that moves around the user's head according to the shooter's aim relative to the target center. As the shooter approaches the target from one direction, the sound source moves in front of them from the opposite direction.

### 4 ONGOING EXPERIMENTS

Both experiments follow the same procedure, differing only in conditions. A CS session has 2 blocks per sonification method (with and without sound center), and a *SPAT* session has 3 blocks per method (no spatialization, binary, 3D continuous), totaling 6 blocks each.

After a familiarization phase with the sonification method and a training phase, the participant is asked to shoot at 4 invisible targets appearing at predetermined positions in front of them, one after the other. Between each shot, the participant can rest and then shoot at a disk to start the next trial.

The experiment is still ongoing, with a total of 24 sighted participants and 8 BVI participants expected to take part. Preliminary results from 6 sighted and 1 visually impaired participant suggest that while the central sound improved shooting precision, it had no significant effect on shooting speed.

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<sup>1</sup><https://tinyurl.com/28wraswn>